

Learning Control

Part II: Optimization-based ILC design

Tom Oomen

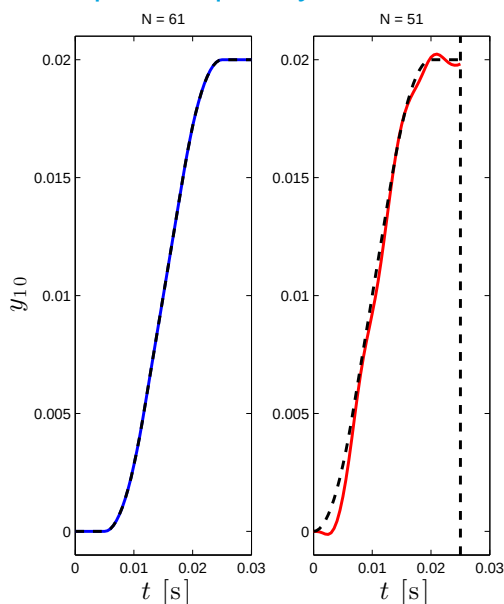
Control Systems Technology Group
Department of Mechanical Engineering
Eindhoven University of Technology

2025/2026

Introduction

2/71

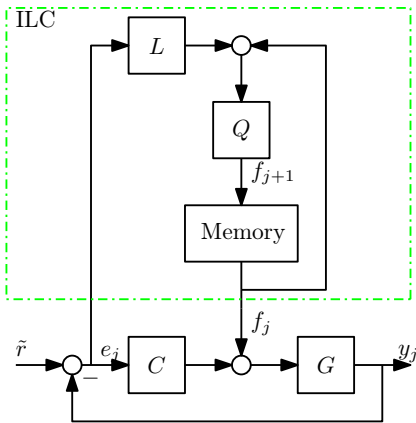
Example: frequency domain ILC



Result

- ▶ $L = (GS)^{-1}$, $Q = 1$
- ▶ faster trajectory
- ▶ 10 iterations
- ▶ trial length N
 - ▶ $N = 61$: expected result
 - ▶ $N = 51$: what's wrong?

ILC scheme



ILC summary

- ▶ learning update: $f_{j+1} = Q(f_j + Le_j)$
- ▶ frequency domain design procedure
 - ▶ $L = (GS)^{-1}$
 - ▶ $|Q(1 - GSL)| < 1, \forall \omega \in [0, 2\pi]$
- ▶ underlying assumptions
 - ▶ infinite time
 - ▶ LTI
 - ▶ SISO (mostly)
- ▶ this part:
 - ▶ finite time
 - ▶ LTI & LTV
 - ▶ optimal
 - ▶ mainly MIMO
 - ▶ basis functions
 - ▶ ...

Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

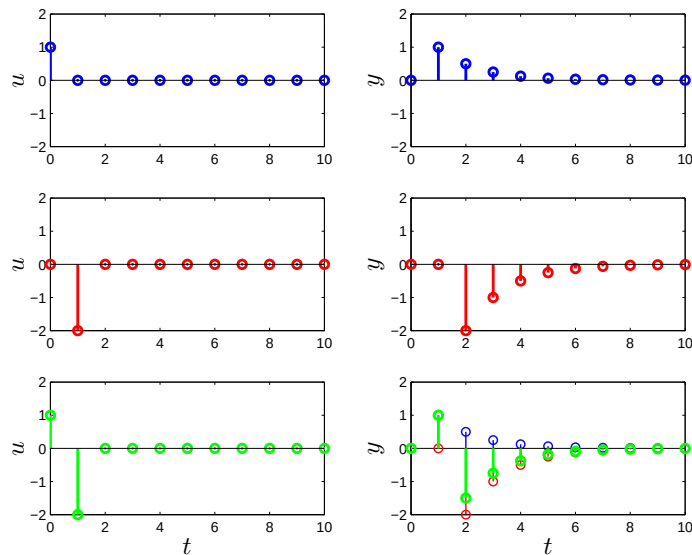
Optimal ILC synthesis

Design Procedure

Summary

Extensions

Intuitive idea



$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ h_{10} \end{bmatrix} u(0)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} 0 \\ h_0 \\ \vdots \\ h_9 \end{bmatrix} u(1)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} h_0 & 0 \\ h_1 & h_0 \\ \vdots & \vdots \\ h_{10} & h_9 \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \end{bmatrix}$$

Convolution causal system

$$y(k) = \sum_{l=0}^{\infty} h_l u(k-l) \quad u(k) = 0, k < 0$$

► $y(0) =$

► $y(1) =$

► $y(2) =$

► $y(3) =$

Convolution matrix

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \underbrace{\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}}_{\text{convolution matrix}} \begin{bmatrix} u(0) \\ u(1) \\ u(2) \\ u(3) \end{bmatrix}$$

- ▶ example: $G = \frac{1}{ms^2}$, $m = 1$
- ▶ $C = 0 \Rightarrow S = 1 \Rightarrow GS = G$
- ▶ differential equation: $\ddot{y} = f$
- ▶ take $x = [y \quad \dot{y}]^T$
- ▶ then state space

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

- ▶ discrete time using $h = 1$ (e.g., c2d.m)

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} f(k)$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(k)$$

- ▶ discrete time using $h = 1$ (e.g., c2d.m)

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} f(k)$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(k)$$

- ▶ impulse response $h_k = \begin{cases} CA^{k-1}B, & k \geq 1 \\ 0 & k \leq 0 \end{cases}$

- ▶ $h_0 = 0$

- ▶ $h_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} = \frac{1}{2}$

- ▶ $h_2 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} = \frac{3}{2}$

- ▶ $h_3 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} = \frac{5}{2}$

► impulse response $h_k = \begin{cases} CA^{k-1}B, & k \geq 1 \\ 0 & k \leq 0 \end{cases}$

► $h_0 = 0$

► $h_1 = [1 \ 0] \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{2}$

► $h_2 = [1 \ 0] \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \frac{3}{2}$

► $h_3 = [1 \ 0] \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \frac{5}{2}$

► convolution matrix $y_j = Jf_j$

$$J = \begin{bmatrix} h_0 & 0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 & 0 \\ h_3 & h_2 & h_1 & h_0 & 0 \\ h_4 & h_3 & h_2 & h_1 & h_0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{3}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{5}{2} & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ \frac{7}{2} & \frac{5}{2} & \frac{3}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

Frequency domain ILC controller

► $f_{j+1} = Q(f_j + Le_j)$

► set $L = (GS)^{-1}$

► set $Q = 1$

Intuitive lifted ILC controller

► $f_{j+1} = f_j + Le_j$

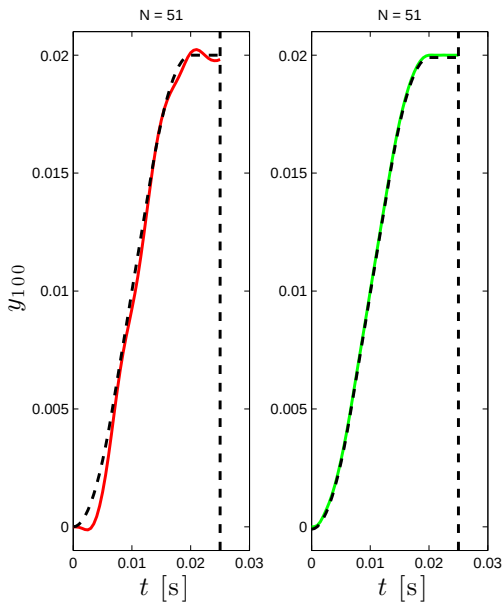
► Tikhonov regularized inverse

► $L = (J^T J + \rho I)^{-1} J^T$

► if J regular and $\rho = 0$, then $L = J^{-1}$

► also known as ridge regression in statistics

Example revisited: frequency domain ILC



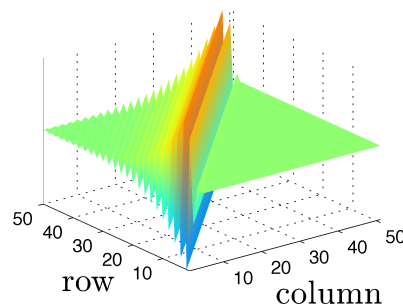
Result after many iterations

- **Frequency domain**: large error
- **lifted ILC**: small error...!

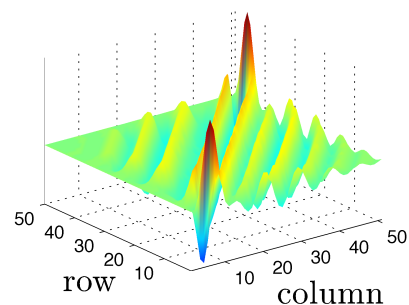
Example revisited: frequency domain ILC

Toeplitz matrix of L :

Frequency domain ILC



Lifted ILC



- Important difference: edge effects in **red** and **blue**
- Note: frequency domain ILC uses ZPETC

Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

Optimal ILC synthesis

Design Procedure

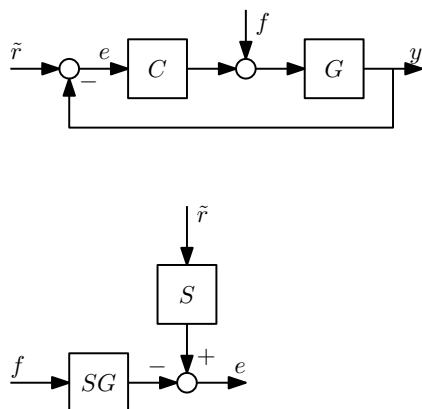
Summary

Extensions

Formal Lifted System Approach

12/71

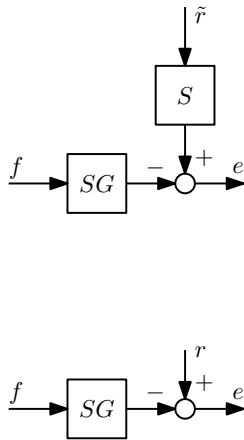
ILC scheme



System formulation

- ▶ back to the original feedback configuration
- ▶ define $S = (I + GC)^{-1}$
- ▶ system relation
 - ▶ $e = S\tilde{r} - SGf$
- ▶ variables
 - ▶ f : control input
 - ▶ $\tilde{r}$: disturbance input
 - ▶ e : output

ILC scheme



Redefine disturbance

- ▶ assume $\tilde{r}(k)$ periodic with period N
- ▶ consider asymptotic situation:

$$r = S\tilde{r} \quad \text{periodic}$$

Introduce state-space realisations

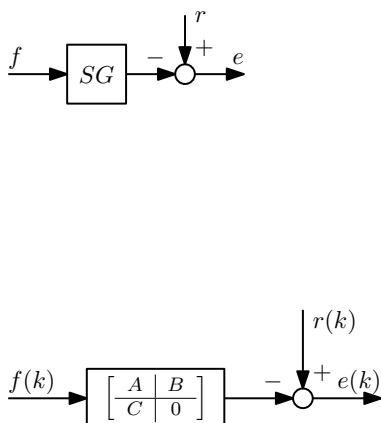
- ▶ minimal state-space realisation for G :

$$\left[\begin{array}{c|c} A_G & B_G \\ \hline C_G & 0 \end{array} \right] \Rightarrow \text{strictly proper}$$

- ▶ minimal state-space realisation for C :

$$\left[\begin{array}{c|c} A_C & B_C \\ \hline C_C & D_C \end{array} \right] \Rightarrow \text{proper}$$

ILC scheme



Resulting closed-loop state-space

$$\begin{aligned} x(k+1) &= Ax(k) + Bf(k) \\ e(k) &= r(k) - Cx(k) \end{aligned}$$

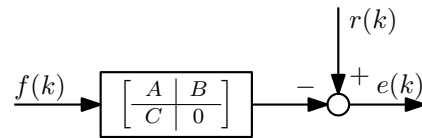
$$\begin{aligned} A &= \begin{bmatrix} A_G - B_G D_C C_G & B_G C_C \\ -B_C C_G & A_C \end{bmatrix} \\ B &= \begin{bmatrix} B_G \\ 0 \end{bmatrix} \quad C = [C_G \quad 0] \end{aligned}$$

General problem setup

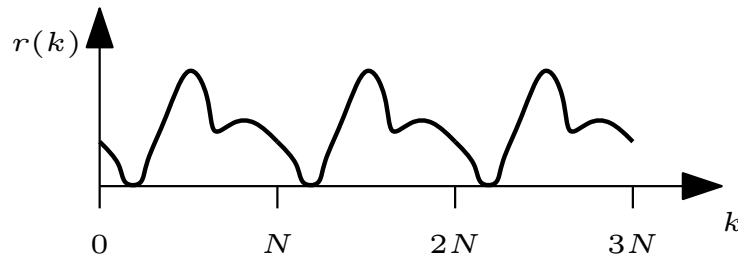
- ▶ repetitive control
- ▶ iterative learning control

General setup: repetitive control

- ▶ periodic disturbance, e.g.,
 - ▶ drive control with nonuniform torque pattern
 - ▶ optical disc eccentricity

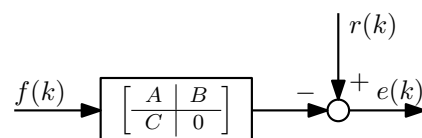


- ▶ $x(0) = 0$, whereas $x(N)$, $x(2N)$, $x(3N)$ depend on algorithm
- ▶ typical disturbance r :

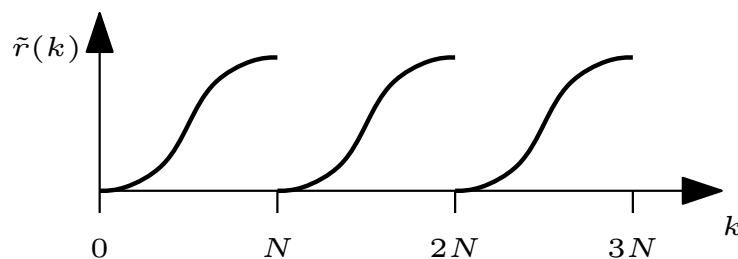


General setup: iterative learning control

- ▶ servo problem, e.g.,
 - ▶ pick and place machine



- ▶ $x(0) = 0$, $x(N) = 0$, $x(2N) = 0$, $x(3N) = 0$
- ▶ typical disturbance $\tilde{r}$:



Intermezzo: state-space models

- ▶ state-space model

$$x(k+1) = Ax(k) + Bf(k)$$

$$y(k) = Cx(k)$$

interpretation: state $x(k)$ contains effect of all inputs up to time k

⇒ computing $y(N)$ requires $x(N)$

- ▶ impulse response model $h_k \in \mathbb{R}^{m \times l}$

$$y(k) = h_k \otimes f(k) = \sum_{l=0}^{\infty} h_l f(k-l), \quad h_k = \begin{cases} CA^{k-1}B, & k \geq 1 \\ 0 & k \leq 0 \end{cases}$$

⇒ computing $y(N)$ requires all past inputs $f(k)$, $k = -\infty, \dots, N$

- ▶ **Key idea**: combine the two concepts:
process data of length $N = 1$ experiment = 1 trial

Trial domain

- ▶ let j denote trial number, with trial length N
- ▶ define for a vector $z(k)$

$$z_j = \begin{bmatrix} z(Nj) \\ z(Nj+1) \\ \vdots \\ z(Nj+N-1) \end{bmatrix}$$

- ▶ thus: $r_j = r, \forall j$
- ▶ **lifted** system representation

$$x(Nj+N) = Fx(Nj) + Gf_j$$

$$y_j = Hx(Nj) + Jf_j$$

$$e_j = r - y_j$$

Trial domain

- **lifted** system representation

$$x(Nj + N) = Fx(Nj) + Gf_j$$

$$y_j = Hx(Nj) + Jf_j$$

$$e_j = r - y_j$$

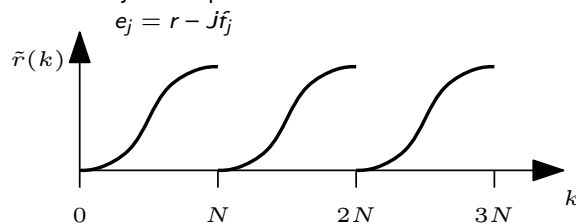
- here:

$$\left[\begin{array}{c|c} F & G \\ \hline H & J \end{array} \right] = \left[\begin{array}{c|cccccc} A^N & A^{N-1}B & A^{N-2}B & \dots & \dots & AB & B \\ \hline C & h_0 & 0 & 0 & \dots & \dots & 0 \\ CA & h_1 & h_0 & 0 & \dots & \dots & 0 \\ CA^2 & h_2 & h_1 & h_0 & \dots & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ CA^{N-1} & h_{N-1} & \dots & \dots & h_2 & h_1 & h_0 \end{array} \right]$$

- similar to lifting for LPTV systems (Bamieh et al. 1991) (Bamieh & Pearson 1992)

Lifted system representation

- convolution matrix J :
 - time invariance: J Toeplitz
 - causality: J lower triangular
- (A, B, C) minimal $\Rightarrow (F, G, H)$ minimal
- ILC case:
 - state reset: $x(0) = 0, x(N) = 0, x(2N) = 0, x(3N) = 0$
 - lifted system representation reduces to



Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

Optimal ILC synthesis

Design Procedure

Summary

Extensions

Internal Model Principle

21/71

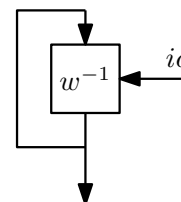
Internal model principle

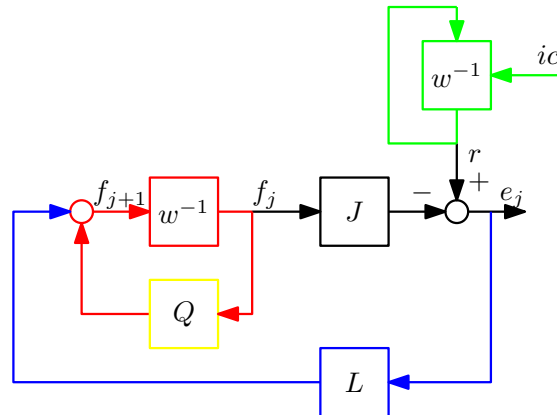
Asymptotic rejection of a constant disturbance r is achieved, i.e., $e_j \rightarrow 0$ for $j \rightarrow \infty$, if

1. error-driven disturbance model is included in controller
2. disturbance can propagate through system, i.e., not canceled by transmission zero
3. dynamics of system and controller are asymptotically stabilized by feedback

Basic idea

Constant disturbance $r \in \mathbb{R}^N$ generated in trial domain by integrator with initial condition $ic = r$





Internal model principle: ILC

- ▶ disturbance model
- ▶ include disturbance model in controller
- ▶ apply feedback L to stabilize
- ▶ optionally include Q for robustness

Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

Optimal ILC synthesis

Design Procedure

Summary

Extensions

System relations

- ▶ system
 - ▶ $e_j = r - Jf_j$
 - ▶ J square, lower triangular and Toeplitz
 - ▶ time invariant
 - ▶ causal
- ▶ ILC controller
 - ▶ $f_{j+1} = Qf_j + Le_j$
 - ▶ Q, L not necessarily lower triangular and Toeplitz
 - ▶ time varying
 - ▶ non-causality
 - ▶ system relations

$$f_{j+1} = (Q - LJ) f_j + Lr$$

$$e_j = r - Jf_j$$

ILC analysis

- ▶ system relations resembles state-space system

$$f_{j+1} = (Q - LJ) f_j + Lr$$

$$e_j = -Jf_j + r$$

- ▶ system is asymptotically stable if $|\lambda_i(Q - LJ)| < 1, \quad i = 1, 2, \dots, Nl$
- ▶ state-space to iteration-domain transfer function matrix $r \mapsto e_j$

$$e_j = \left(I_{Nl} - J(wI_{Nl} - Q + LJ)^{-1} L \right) r$$

- ▶ steady-state error: set $w = 1$

$$e_\infty = \left(I_{Nl} - J(I_{Nl} - Q + LJ)^{-1} L \right) r$$

Perfect ILC performance

- ▶ steady state error: $e_\infty = (I_{NI} - J(I_{NI} - Q + LJ)^{-1}L) r$
- ▶ let $Q = I_{NI}$. Then $e_\infty = (I_{NI} - J(LJ)^{-1}L) r$
- ▶ $e_\infty = 0$ requires LJ to be non-singular
 $\Rightarrow J$ and L square and non-singular

Singularity of J

- ▶ reasons:
 - ▶ strictly proper systems or additional delays:
 if $h_0, h_1, \dots, h_{d-1} = 0$, then $\text{rank}(J)$ is at most $Nm - d$
 - ▶ non-minimum phase zeros
- ▶ consequences:
 - ▶ no full stabilization
 - ▶ no full rejection

Example 1

$$h = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \Rightarrow J = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 0 \end{bmatrix}, \quad r = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Hence

$$\text{Im} J = \text{Im} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ -1 & 1 \end{bmatrix} = \text{Im} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- ▶ J is output matrix in system with unit system matrix
- ▶ third mode is **unobservable**
- ▶ feedback has no effect on pole location of third mode
- ▶ $e_\infty(1) = 1$

Example 2

$$H(z) = \frac{z-2}{z(z+1)} = -\frac{2}{z} + \frac{3}{z+1}$$

$$\left[\begin{array}{c|c} zI - A & B \\ \hline -C & \end{array} \right] = \left[\begin{array}{cc|c} z & 0 & -2 \\ 0 & z+1 & 3 \\ \hline -1 & -1 & \end{array} \right]$$

zero at $z = 2$

impulse response: $h = \begin{bmatrix} 0 \\ 1 \\ -3 \\ 3 \\ -3 \end{bmatrix}$, observability matrix: $\begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 0 & 1 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}$

Example 2

$$J = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 & 0 \\ 3 & -3 & 1 & 0 & 0 \\ -3 & 3 & -3 & 1 & 0 \end{bmatrix}$$

By definition of zeros there exists x_0 and exponential input signal containing zero as exponential factor, such that resulting output is zero:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 & 0 \\ 3 & -3 & 1 & 0 & 0 \\ -3 & 3 & -3 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 0 & 1 \\ 0 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = 0$$

Example 2

or $Jf_j + Hx_0 = 0$

Then upper bound on smallest non-zero singular value of J is given by

$$\underline{\sigma}(J) < \frac{\|Hx_0\|}{\|f_j\|}$$

For non-minimum phase zero $\|f_j\|$ will grow for larger dimensions N forcing $\underline{\sigma}(J)$ to be small

Summary examples

- ▶ delays
- ▶ non-minimum phase zeros

lead to loss of rank J

Convergence properties revisited

- ▶ $e_j = r - Jf_j$
- ▶ $f_{j+1} = Qf_j + Le_j = (Q - LJ)f_j + Lr$
- ▶ asymptotically stable if $|\lambda_i(Q - LJ)| < 1, \quad i = 1, 2, \dots, N$

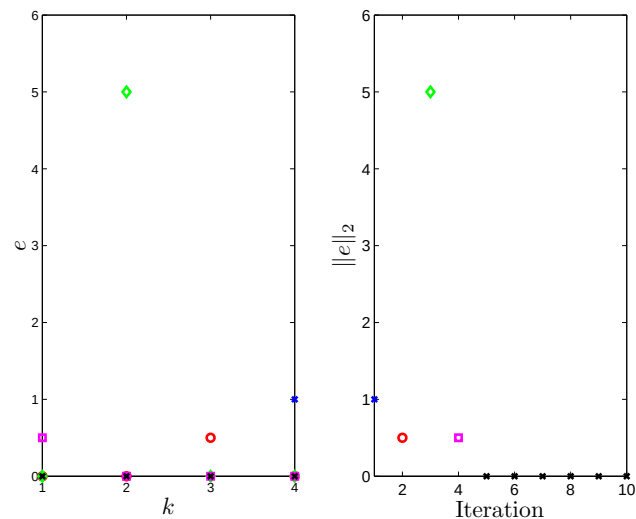
Example

$$J = I_4, \quad Q = I_4, \quad L = \begin{bmatrix} 1 & -\frac{1}{10} & 0 & 0 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ▶ Q : asymptotically stable?

Example

- ▶ $f_1 = 0$
- ▶ $e_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$
- ▶ error **increases** at iteration 3
- ▶ but indeed **converges** to $e_\infty = 0$
- ▶ **undesired** transient behavior



ILC learning transient

- ▶ no guarantees provided by convergence condition $\lambda_i(Q - LJ)$
- ▶ of essential importance in practice

Result

Monotonic convergence of f_j in the ℓ_2 norm, i.e.,

$$\|f_\infty - f_{j+1}\| \leq k \|f_\infty - f_j\|$$

for some $k \in [0, 1)$, is achieved if

$$\|Q - LJ\| < 1$$

Note: can also be defined for e_j

Convergence conditions

- ▶ convergence: $\max_i |\lambda_i(Q - LJ)| < 1$
- ▶ monotonic convergence: $\|Q - LJ\| < 1$

Convergence results

- ▶ spectral radius $\max_i |\lambda_i(Q - LJ)|$ is not a norm
- ▶ for any (matrix) norm $\|\cdot\|$: $\max_i |\lambda_i(Q - LJ)| \leq \|Q - LJ\|$
- ▶ hence: monotonic convergence $\Rightarrow$ convergence
- ▶ monotonic convergence in 2-norm obtained if $\bar{\sigma}(Q - LJ) < 1$
- ▶ convergence in previous part (frequency domain): also **monotonic**

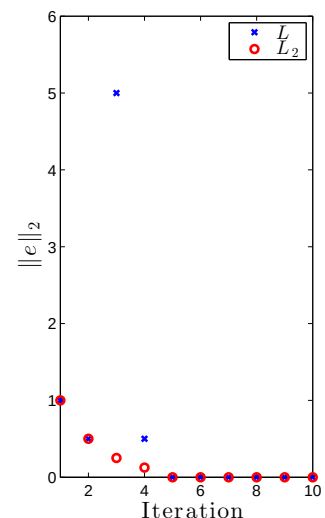
Example revisited

▶ $L = \begin{bmatrix} 1 & -\frac{1}{10} & 0 & 0 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ▶ $\max_i |\lambda_i(Q - LJ)|$
- ▶ $\bar{\sigma}(Q - LJ)$

▶ $L_2 = \begin{bmatrix} 1 & -\frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ▶ $\max_i |\lambda_i(Q - L_2J)|$
- ▶ $\bar{\sigma}(Q - L_2J)$



Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

Optimal ILC synthesis

Design Procedure

Summary

Extensions

Optimal ILC synthesis

35/71

- ▶ system relations

$$\begin{aligned}f_{j+1} &= Qf_j + Le_j \\ e_j &= r - Jf_j\end{aligned}$$

- ▶ how to design Q and L ?
- ▶ formalization of Tikhonov regularization on slide 9
- ▶ consider criterion

$$\mathcal{J}(f_{j+1}) = e_{j+1}^T W_e e_{j+1} + f_{j+1}^T W_f f_{j+1} + (f_{j+1} - f_j)^T W_{\Delta f} (f_{j+1} - f_j)$$

- ▶ goal: compute $\arg \min_{f_{j+1}} \mathcal{J}(f_{j+1})$
- ▶ weights $W_e, W_f, W_{\Delta f}$:
 - ▶ symmetric, positive (semi) definite
 - ▶ typical choice: $W_e = I, W_{\Delta f} = w_{\Delta f} I, W_f = w_f I$, with $w_f, w_{\Delta f} \geq 0$

Reformulate

$$e_j = r - Jf_j \Rightarrow r = e_j + Jf_j$$

Hence

$$\begin{aligned} e_{j+1} &= r - Jf_{j+1} \\ &= e_j - J(f_{j+1} - f_j) \end{aligned}$$

and

$$\begin{aligned} \mathcal{J}(f_{j+1}) &= (e_j - J(f_{j+1} - f_j))^T W_e (e_j - J(f_{j+1} - f_j)) \\ &\quad + f_{j+1}^T W_f f_{j+1} + (f_{j+1} - f_j)^T W_{\Delta f} (f_{j+1} - f_j) \end{aligned}$$

Optimal ILC

- first order necessary condition for optimality: $\frac{\partial \mathcal{J}(f_{j+1})}{\partial f_{j+1}} = 0$

Result

- use identity $\frac{\partial (s-Ax)^T W (s-Ax)}{\partial x} = -2A^T W (s-Ax)$

- then $\frac{\partial \mathcal{J}(f_{j+1})}{\partial f_{j+1}} = 0 \Rightarrow$ implies

$$(J^T W_e J + W_f + W_{\Delta f}) f_{j+1} - (J^T W_e J + W_{\Delta f}) f_j - J^T W_e e_j = 0$$

$$f_{j+1} = \underbrace{(J^T W_e J + W_f + W_{\Delta f})^{-1} (J^T W_e J + W_{\Delta f})}_Q f_j + \underbrace{(J^T W_e J + W_f + W_{\Delta f})^{-1} J^T W_e e_j}_L$$

Analysis of optimal ILC

- monotonic convergence:

$$\bar{\sigma}(Q - LJ) = \bar{\sigma} \left((J^T W_e J + W_f + W_{\Delta f})^{-1} W_{\Delta f} \right) < 1$$

- result: since $J^T W_e J + W_f \succeq 0$

1. if $J^T W_e J + W_f \succ 0$, i.e., full rank, then guaranteed monotonic convergence
2. if J singular, then set $W_f \succ 0$ to guarantee monotonic convergence
3. if $W_{\Delta f} = 0$, then deadbeat ILC
4. if J non-singular, then $W_f = W_{\Delta f} = 0$ and $W_e = w_e I$ leads to

$$f_{j+1} = f_j + (J^T J)^{-1} J^T e_j = f_j + J^{-1} e_j \Rightarrow \text{inverse model ILC}$$

Steady state error

$$e_{\infty} = \left(I_{Nl} - J(I_{Nl} - Q + LJ)^{-1} L \right) r$$

For optimal ILC:

$$e_{\infty} = \left(I_{Nl} - J(J^T W_e J + W_f)^{-1} J^T W_e \right) r$$

Design guidelines

- if J non-singular, then $W_f = 0$ leads to $e_{\infty} = 0$
- can be shown that $W_f \succ 0$ leads to improved robustness w.r.t. model uncertainty
- $W_{\Delta f}$:
 - coincides with specific Tikhonov regularization on slide 9
 - large: attenuate trial-variant disturbances
 - small: faster convergence in trial-invariant disturbance situation

Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

Optimal ILC synthesis

Design Procedure

Summary

Extensions

Design Procedure

40/71

1. Obtain an impulse response of system
 - ▶ from a given parametric model
 - ▶ from experiments
 - ▶ impulse response testing
 - ▶ estimation of FIR model using general inputs (fairly easy)
2. Compute Toeplitz matrix J
3. Select tuning parameters W_e , W_f , and $W_{\Delta f}$
4. Compute optimal ILC: L and Q on slide 37

Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

Optimal ILC synthesis

Design Procedure

Summary

Extensions

Summary

41/71

Lifted ILC

- ▶ finite time, LTI/LTV, MIMO
- ▶ general framework for repetitive control and ILC (de Roover et al. 2000)

Analysis using linear algebra

- ▶ convergence vs. monotonic convergence

Synthesis

- ▶ optimal ILC algorithms

Extensions

- ▶ many different extensions and applications
 - ▶ see overview papers, e.g., (Bristow et al. 2006), (Blanken, de Rozario, van Zundert, Koekebakker, Steinbuch & Oomen 2016)
 - ▶ many contributions of CST M.Sc./Ph.D. students, e.g., <http://www.toomen.eu>

Basic Finite Time ILC Framework

Formal Lifted System Approach

Internal Model Principle

ILC analysis

Optimal ILC synthesis

Design Procedure

Summary

Extensions

Extensions: Low-Order ILC

42/71

Low-order implementation of optimal ILC controller of slide 37 (Oomen et al. 2011)

(van Zundert et al. 2016)

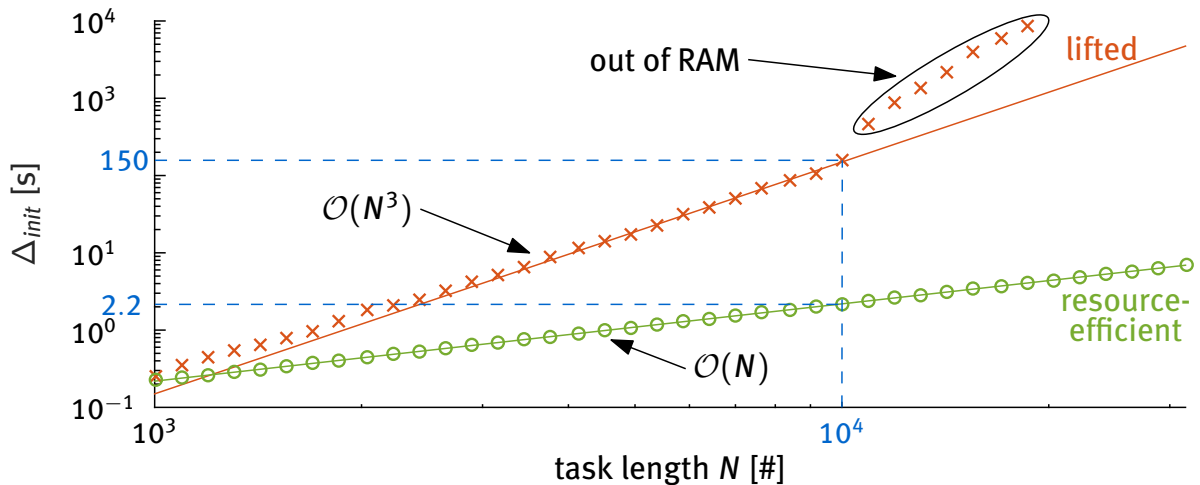
$$f_{j+1} = C^{\text{ILC}}(k) \begin{bmatrix} e_j(t) \\ f_j(k) \\ g_j(k+1) \end{bmatrix}$$

- ▶ state-space $C^{\text{ILC}}(k)$ follows from Riccati difference equation
- ▶ related to LQ tracking problem: $g_{<k>}(k+1)$ noncausal
- ▶ **tractable** for large scale problems
 - ▶ matrix dimensions invariant under
 - ▶ intersample behavior
 - ▶ $\#n_u, \#n_y$: MIMO
 - ▶ N : trial length
 - ▶ **in contrast**: approach on slide 37 **inflates** matrix dimensions

Low-order implementation of optimal ILC controller of slide 37 (Oomen et al. 2011)

(van Zundert et al. 2016)

Large computational improvements



Note: different ILC implementation with identical control performance

Extensions: Position-dependent & LTV systems

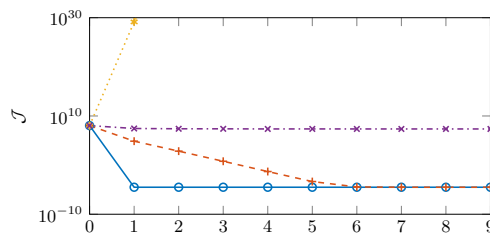
Many systems are time-varying

- ▶ state-space: $(A_0(k), B_0(k), C_0(k), D_0(k))$
- ▶ examples
 - ▶ moving mass stages: commutation $B_0(k)$, varying sensor measurement: varying $C_0(k)$
 - ▶ H-bridge stages: also varying $A_0(k)$
 - ▶ nuclear fusion: varying $A_0(k), B_0(k), C_0(k), D_0(k)$
- ▶ learning requires inverse: state-space given by $(A_0(k) - B_0(k)D_0^{-1}(k)C_0(k), B_0(k)D_0^{-1}(k), -D_0^{-1}(k)C_0(k), D_0^{-1}(k))$
- ▶ complex dependence on original system matrices
- ▶ **good news**: without increasing computational requirements
 - ▶ stable inversion (see previous part - frequency domain ILC) applies
 - ▶ optimal ILC of slide 37 (and slide 42) directly applies
 - ▶ many more variants...
- ▶ next: application to a position-dependent system

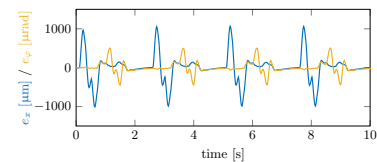
ILC for position-dependent motion feedforward

(van Zundert et al. 2016)

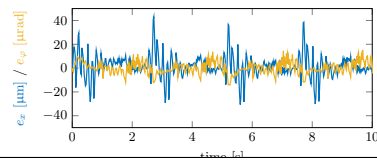
- ▶ position-dependent system
- ▶ position-dependent/LTV ILC for fast learning
- ▶ LTI ILC: models at various locations:
 - ▶ LTI ILC: convergent
 - ▶ LTI ILC: nonconvergent
 - ▶ LTI ILC: convergent with Q-filter
- ▶ learning leads to extreme accuracy!



Before learning:



Using ILC convergence:

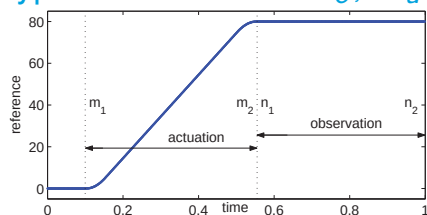


Extensions: Hankel ILC

Time-varying weighting filters: Hankel ILC (van de Wijdeven & Bosgra 2008) (van de Wijdeven 2008)

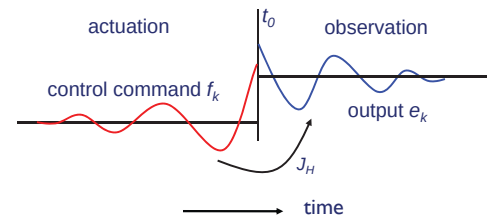
- ▶ set $W_o = \begin{bmatrix} 0_{n \times n_1} & I_n & 0_{n \times (N-n_2-1)} \end{bmatrix}$
- ▶ set $W_a = \begin{bmatrix} 0_{m_1 \times m} \\ I_m \\ 0_{(N-m_2-1) \times m} \end{bmatrix}$

Typical selection of W_o, W_a



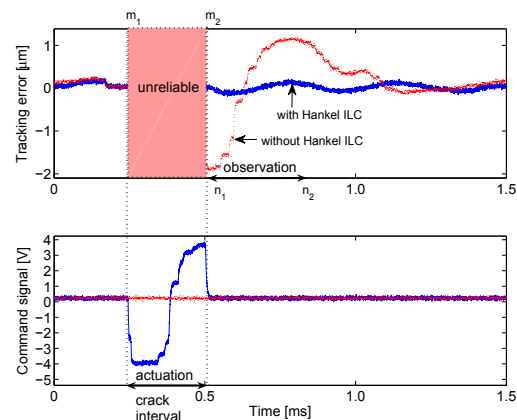
Connection to Hankel matrix

- ▶ $J_H = W_o J W_a$

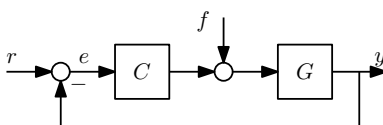


Hankel ILC applications

- ▶ residual vibrations in point-to-point tasks (van de Wijdeven & Bosgra 2008)
- ▶ reading cracked optical discs (Steinbuch, van de Wijdeven, Oomen, van Berkel & Leenknecht 2009)
(Steinbuch, van Berkel, Leenknecht, Oomen & van de Wijdeven 2009)

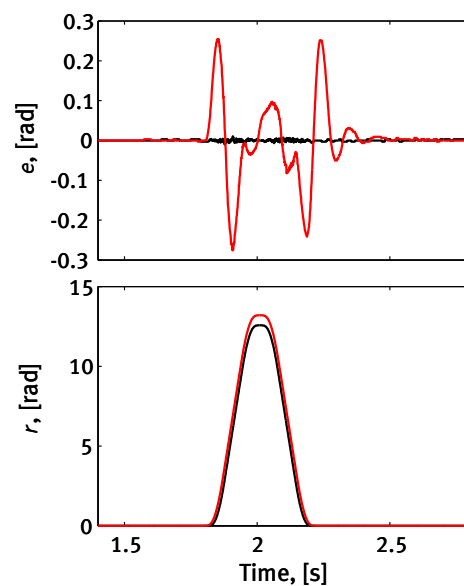


Extensions: ILC with basis functions

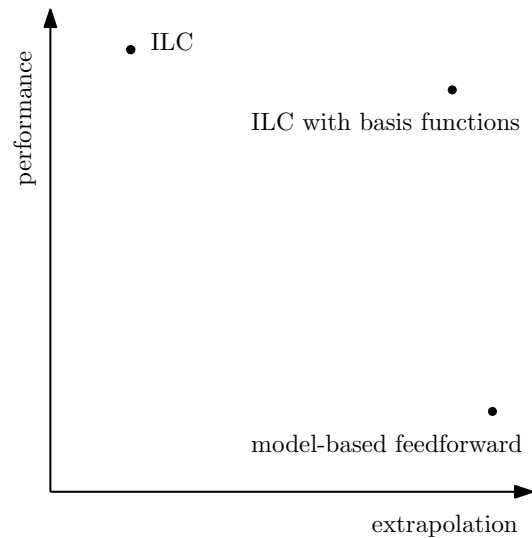


Iteration-varying reference r

- ▶ Learned ILC signal (black)
 - ▶ $e_1 = S(r_1 - Gf_1) \approx 0$
- ▶ change of r at experiment 4 (red)
 - ▶ $e_4 = S(r_4 - Gf_1)$: large!



Extrapolation vs. performance?

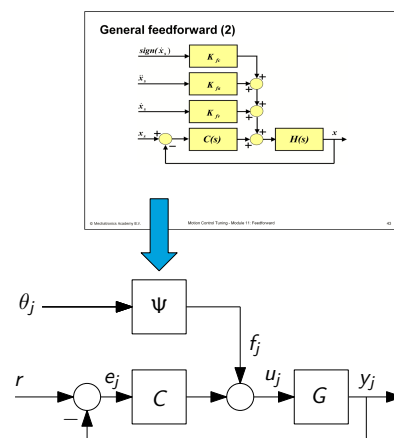


Extensions: ILC with basis functions

ILC with basis functions

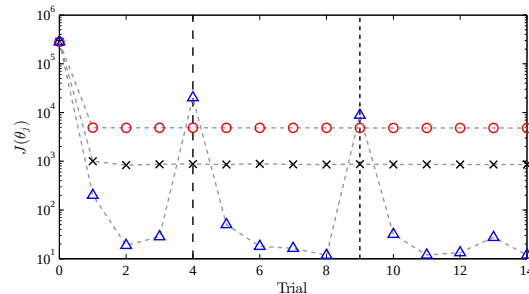
- ▶ parameterise: $f_j(\theta_j, r)$
 - ▶ e.g., $f_j = \Psi(r)\theta_j$
- ▶ then, in finite time notation

$$e_j = S(r - G\Psi(r)\theta_j)$$
- ▶ if $G\Psi(r)\theta_j \approx r$, then good extrapolation
- ▶ example: $G = \frac{1}{ms^2}$. then $\Psi(r) = \ddot{r}$
 $\Rightarrow$ autotuning mass feedforward!



ILC with basis functions: change of r at iteration 4 & 9

- ▶ **blue**: ILC of slide 37
- ▶ **red**: polynomial basis (van de Wijdeven & Bosgra 2010) (van der Meulen et al. 2008) (Bolder et al. 2012)
- ▶ **black**: rational basis (Bolder & Oomen 2015)



- ▶ basis functions lead to enhance flexibility for varying tasks
- ▶ but go at the expense of performance for trial-invariant tasks

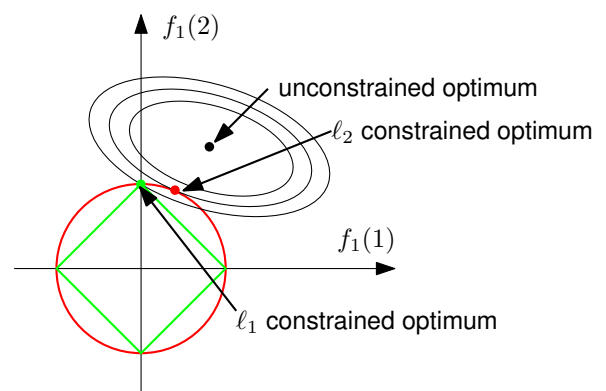
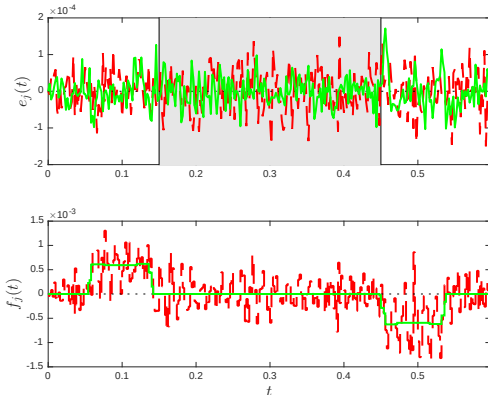
Basis functions in ILC - further reading

- ▶ combine advantages learning (performance) and flexibility (model-based feedforward) (Bolder, Boeren & Oomen 2014)
- ▶ norm-optimal framework for rational bases in ILC: (Bolder & Oomen 2015)
- ▶ special case: polynomial bases: (van de Wijdeven & Bosgra 2010) (van der Meulen et al. 2008) (Bolder et al. 2012), and input shaping: (Boeren, Bruijnen, van Dijk & Oomen 2014)
- ▶ rational bases: similar implementation issues as L -filter in frequency domain ILC, see also (van Zundert et al. 2016)
- ▶ also identification-based techniques (no L filter required), see (Boeren, Oomen & Steinbuch 2015) (polynomial, nonoptimal), (Boeren, Oomen & Steinbuch 2014) (polynomial, optimal), (Boeren, Blanken, Bruijnen & Oomen 2015) (rational), (Blanken, Boeren, Bruijnen & Oomen 2016) (connection to ILC)
- ▶ frequency domain ILC design with bases functions: (Boeren et al. 2016)
- ▶ feedforward for position-dependent systems (see Slide 44) in (Smits 2016) and associated modelling in (de Rozario 2015)

How to pick the basis functions and order?

- ▶ Lasso for sparse ILC (Oomen & Rojas 2017)
 - ▶ automatic selection of basis functions
 - ▶ noise attenuation
 - ▶ resource efficiency

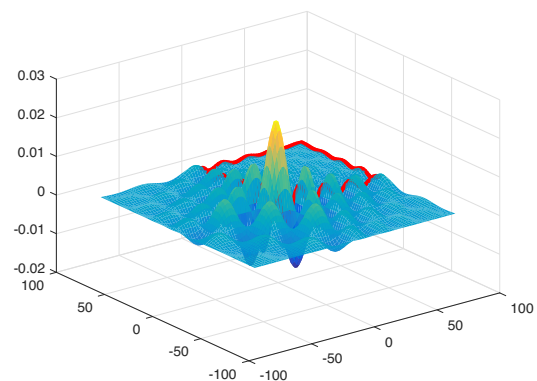
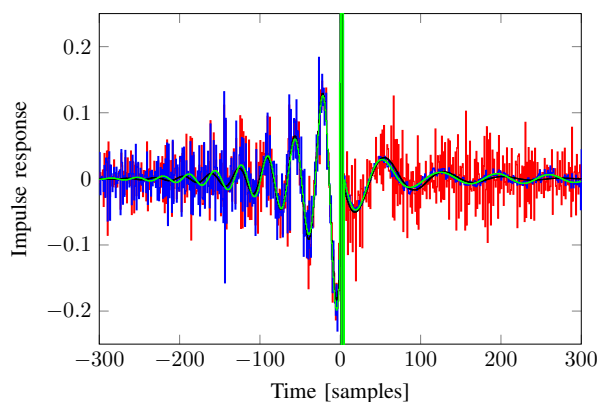
Results: traditional ILC vs. sparse ILC



A different perspective on picking the basis functions and order

- ▶ last decade in SysId and Machine Learning: kernel-based regularization
- ▶ for non-causal systems: [Blanken, PhD 2019]
- ▶ and beyond

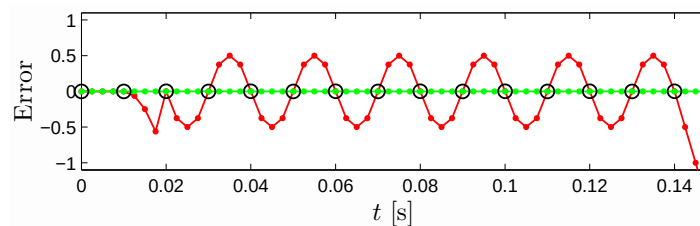
Feedforward results: traditional FIR vs. causal regularized vs. noncausal regularized



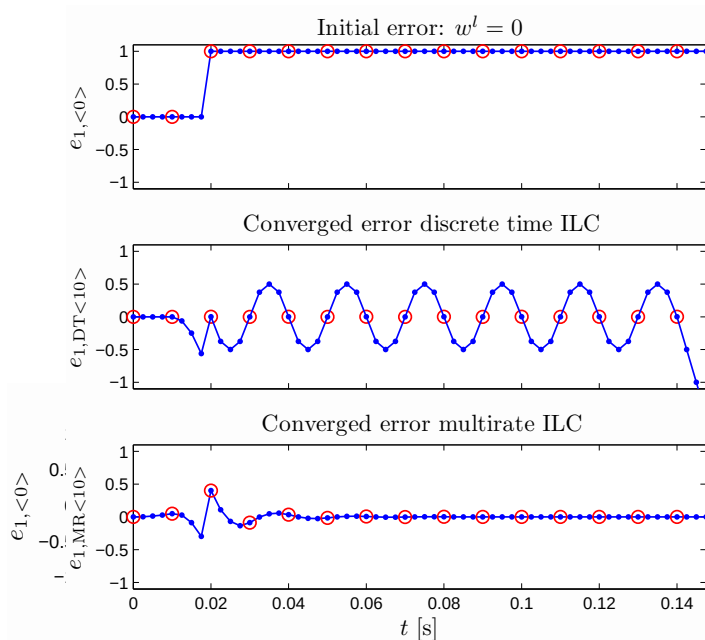
Iterative Learning Control (ILC):

- ▶ control strategy for batch repetitive processes:
- ▶ digital computer implementation:
 - ▶ results in high **discrete time** performance
 - ▶ however: system evolves in **continuous time**
 - ▶ continuous time performance = discrete time performance + intersample behavior

High discrete time performance:
necessary but **not sufficient** for high continuous time performance



Extensions: Sampled-Data ILC



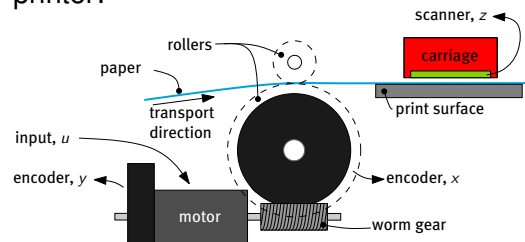
Initial error

Discrete time ILC:
 at-sample: **perfect!**
 Intersample: **poor**

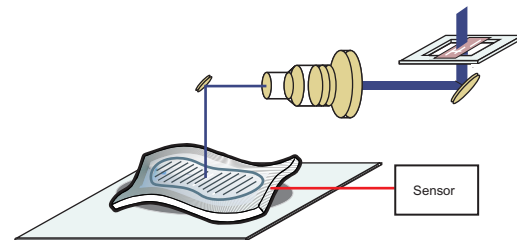
Multirate ILC (Oomen et al. 2009):
 at-sample $\approx$ intersample
 both **good!**

Unmeasured performance variables

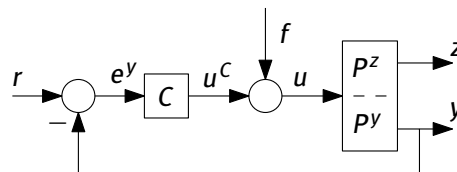
Medium positioning driver in a wide-format printer:

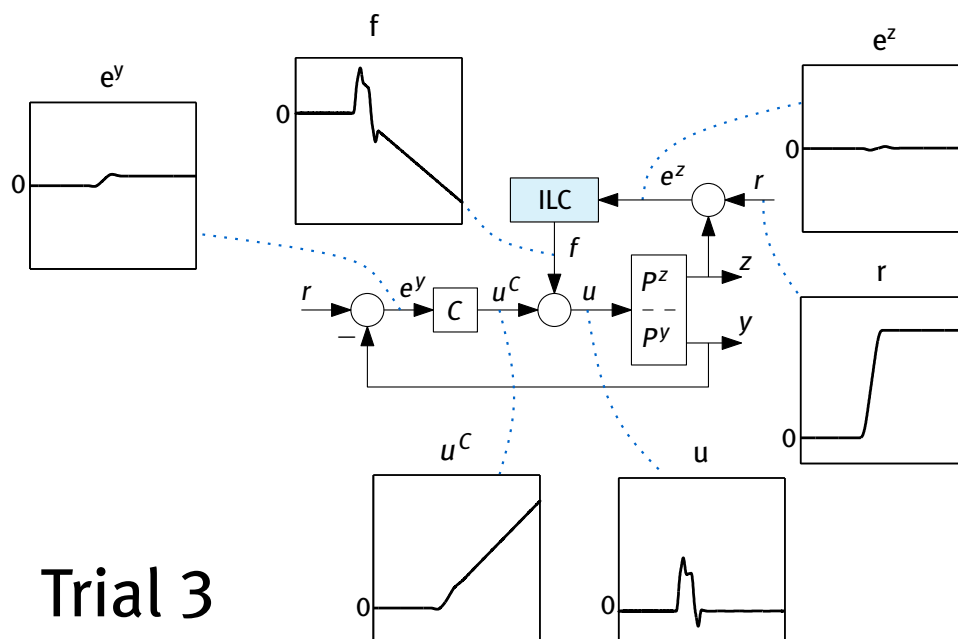
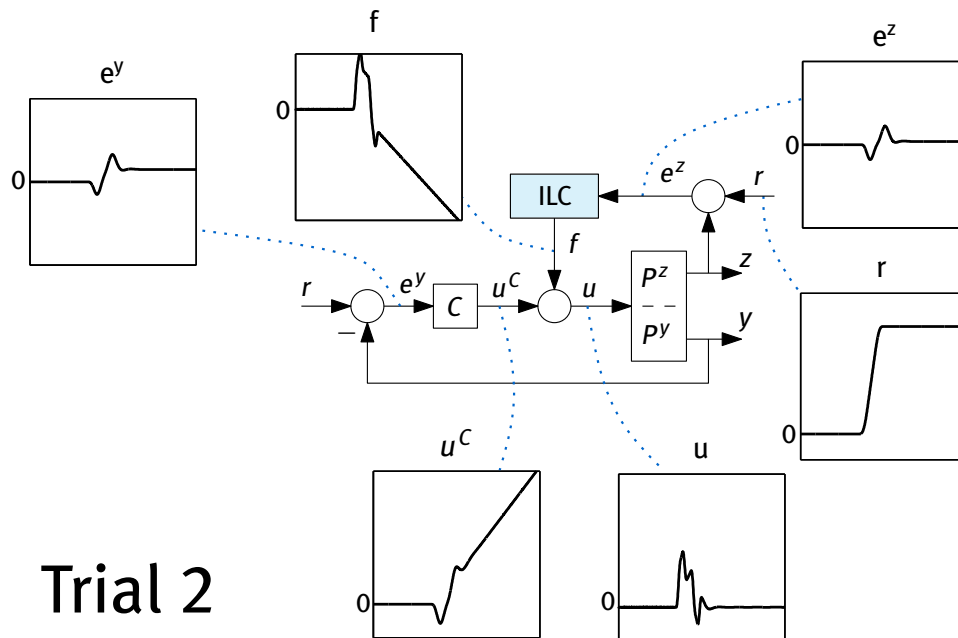


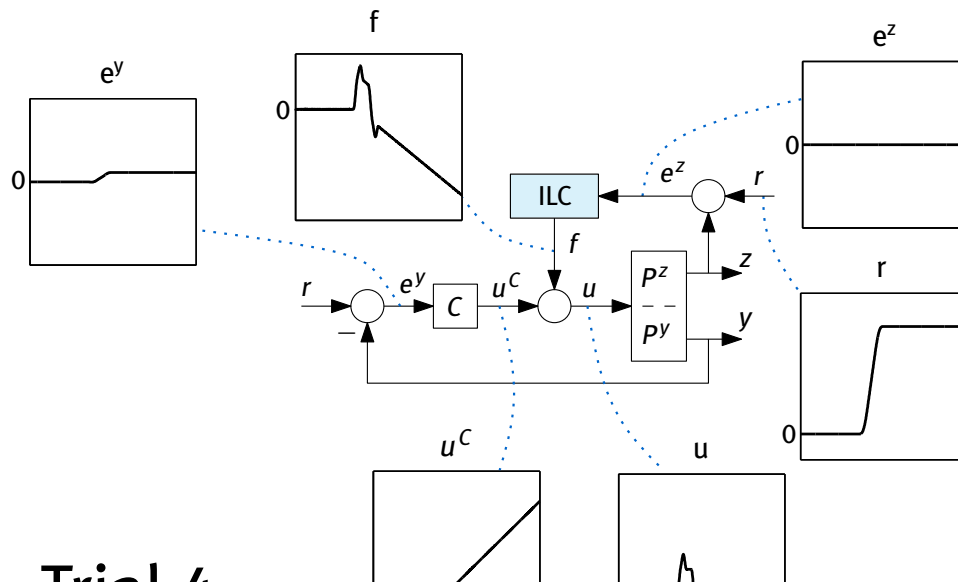
Flexible wafer stage



- real-time measurement y
- performance z , offline available, batch-to-batch
- inferential motion control problem (Oomen et al. 2015)
- inferential ILC: (Bolder & Oomen 2016), see also (Bolder, Oomen & Steinbuch 2014)







Trial 4

This configuration can be internally unstable!

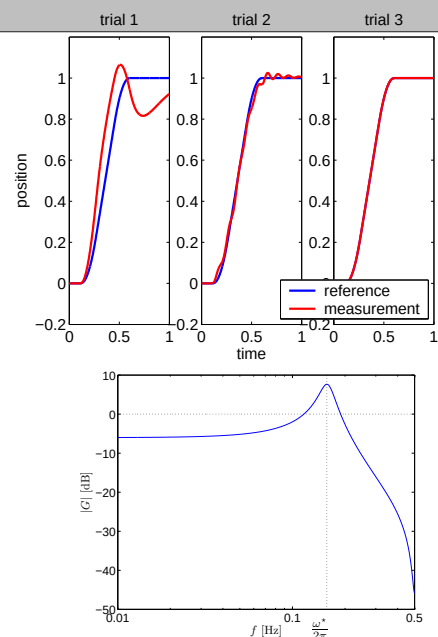
Standard ILC

- ▶ iteratively learn from measured errors
- ▶ goal: reducing error signal
- ▶ extensive theory & applications

Here:

iteratively learn $\mathcal{H}_\infty$ -norm

- ▶ multivariable extensions: (Oomen et al. 2014)
- ▶ application to robust control: (Oomen et al. 2014)
- ▶ noise analysis in (Rojas et al. 2012)





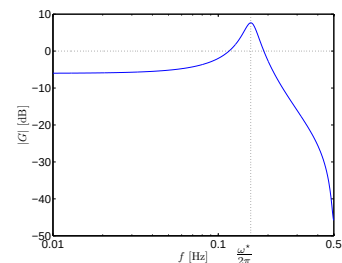
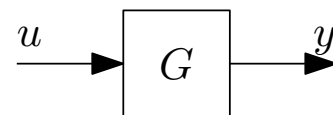
Motivation: robust control

- ▶ model uncertainty bound γ
- ▶ if $\|\Delta_o\|_\infty \leq \gamma$: **guarantees!**
- ▶ if $\|\Delta_o\|_\infty > \gamma$: **no guarantees**
- ▶ if $\|\Delta_o\|_\infty \ll \gamma$: **conservative**

Need accurate bound $\|\Delta_o\|_\infty$

SISO case

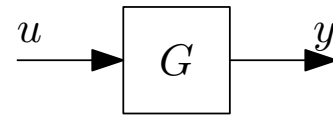
- ▶ $G(z)$ SISO, LTI, stable
- ▶ frequency domain interpretation:
 $\|G\|_\infty = \sup_\omega |G(e^{j\omega})|$



Classical model-based approaches

- ▶ identify parametric model $G(\theta)$
 $\Rightarrow$ compute $\|G(\theta)\|_\infty \Rightarrow$ **model quality?**
- ▶ identify non-parametric frequency response function
 $\Rightarrow$ **intergrid error**

Do we need the auxiliary model $G(\theta)$?



► infinite time: $\|G\|_\infty = \sup_{u \in \ell_2} \frac{\|y\|_2}{\|u\|_2}$

► finite time:

► experiment length N :

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}}_{=y} = \underbrace{\begin{bmatrix} g_0 & 0 & \cdots & 0 \\ g_1 & g_0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ g_{N-1} & g_{N-2} & \cdots & g_0 \end{bmatrix}}_{=G} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix}}_{=u}$$

► then: $\sup_{u \in \ell_2} \frac{\|y\|_2}{\|u\|_2} = \sup_{u \in \ell_2} \sqrt{\frac{y^T y}{u^T u}} = \sup_{u \in \ell_2} \sqrt{\frac{u^T G^T G u}{u^T u}}$

► maximal if u corresponds to the eigenvector $\bar{u}$ of $\lambda^{\max}(G^T G)$:

$$\sup_{u \in \ell_2} \frac{\|y\|_2}{\|u\|_2} = \sqrt{\frac{\bar{u}^T G^T G \bar{u}}{\bar{u}^T \bar{u}}} = \sqrt{\lambda^{\max}(G^T G)}$$

$\Rightarrow$ connection $\|G\|_\infty$ and $\lambda^{\max}(G^T G)$

Analysis of $G^T G$

► exploit Toeplitz structure:

$$G^T = \begin{bmatrix} g_0 & g_1 & \cdots & g_{N-1} \\ 0 & g_0 & \cdots & g_{N-2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & g_0 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{=\mathcal{T} \text{ (time reversal)}} \underbrace{\begin{bmatrix} g_0 & 0 & \cdots & 0 \\ g_1 & g_0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ g_{N-1} & g_{N-2} & \cdots & g_0 \end{bmatrix}}_{=G} \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{=\mathcal{T}}$$

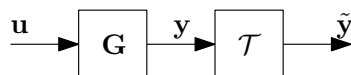
► hence: $G^T G = (\mathcal{T} G)(\mathcal{T} G)$

► Hankel structure: $\mathcal{T} G = (\mathcal{T} G)^T \Rightarrow |\lambda^{\max}(\mathcal{T} G)| = \sqrt{\lambda^{\max}(G^T G)}$

► thus: $|\lambda^{\max}(\mathcal{T} G)| \rightarrow \|G\|_\infty$ for $N \rightarrow \infty$

► remark: $\lambda^{\max}(G) = g_0$

Resulting matrix $\mathcal{T} G$:



We have: $|\lambda^{\max}(\mathcal{T}\mathbf{G})| \rightarrow \|\mathbf{G}\|_\infty$ for $N \rightarrow \infty$

- Given $\bar{\mathbf{u}}$ of $\lambda^{\max}(\mathcal{T}\mathbf{G})$, with $\|\bar{\mathbf{u}}\|_2 = 1$. Evaluate:

$$\bar{\mathbf{u}} \rightarrow \boxed{\mathbf{G}} \rightarrow \bar{\mathbf{y}} \rightarrow \boxed{\mathcal{T}} \rightarrow \tilde{\mathbf{y}} \Rightarrow \lambda^{\max}(\mathcal{T}\mathbf{G}) = \bar{\mathbf{u}}^T \tilde{\mathbf{y}}$$

- **Key problem:** how to obtain $\bar{\mathbf{u}}$? $\Rightarrow$ **power iteration (linear algebra):**

$$\mathbf{u}^{(0)} \rightarrow \boxed{\mathbf{G}} \rightarrow \mathbf{y}^{(0)} \rightarrow \boxed{\mathcal{T}} \rightarrow \tilde{\mathbf{y}}^{(0)} \rightarrow \boxed{\frac{1}{\mu^{(0)}}} \rightarrow \mathbf{u}^{(1)} \rightarrow \dots \rightarrow \mathbf{u}^{(k)} \rightarrow \boxed{\mathbf{G}} \rightarrow \mathbf{y}^{(k)} \rightarrow \boxed{\mathcal{T}} \rightarrow \tilde{\mathbf{y}}^{(k)} \rightarrow \boxed{\frac{1}{\mu^{(k)}}} \rightarrow \mathbf{u}^{(k+1)}$$

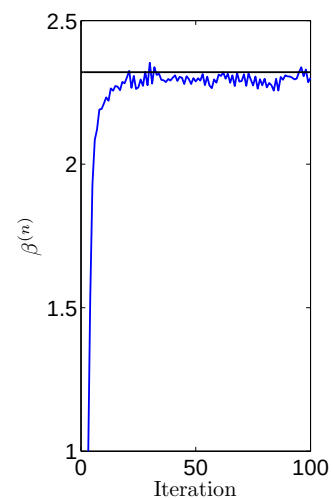
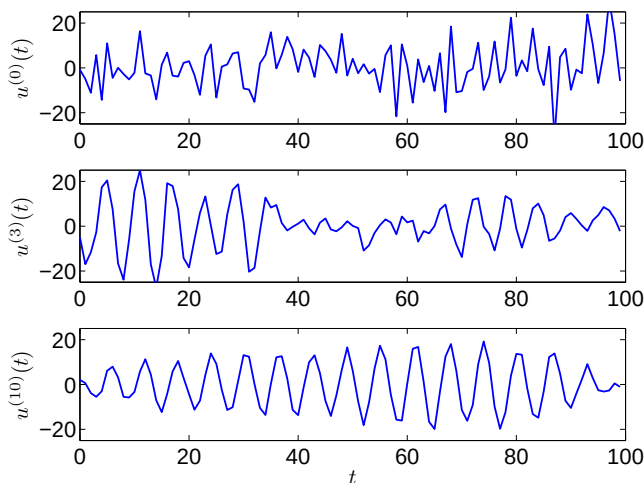
Result: $\mathbf{u}^{(n)} \rightarrow \bar{\mathbf{u}}$ for $n \rightarrow \infty$ (global convergence)

Iteratively learning the $\mathcal{H}_\infty$ -norm (SISO)

- replace matrix $\mathbf{G}$ by evaluation on the true system
- then, $\beta^{(n)} = |(\mathbf{u}^{(n)})^T \tilde{\mathbf{y}}^{(n)}| \rightarrow \|\mathbf{G}\|_\infty$ for $n \rightarrow \infty$

$$\mathbf{u}^{(0)} \rightarrow \boxed{\mathbf{G}} \rightarrow \mathbf{y}^{(0)} \rightarrow \boxed{\mathcal{T}} \rightarrow \tilde{\mathbf{y}}^{(0)} \rightarrow \boxed{\frac{1}{\mu^{(0)}}} \rightarrow \mathbf{u}^{(1)} \rightarrow \dots \rightarrow \mathbf{u}^{(k)} \rightarrow \boxed{\mathbf{G}} \rightarrow \mathbf{y}^{(k)} \rightarrow \boxed{\mathcal{T}} \rightarrow \tilde{\mathbf{y}}^{(k)} \rightarrow \boxed{\frac{1}{\mu^{(k)}}} \rightarrow \mathbf{u}^{(k+1)}$$

Example: iterative procedure converges to $\mathcal{H}_\infty$ norm



Fully data-driven learning

► Data driven feedforward approaches

1. Instrumental variable / correlation-based tuning
(Boeren, Oomen & Steinbuch 2015)
2. ILC without L and Q for MIMO
(Bolder et al. 2018)

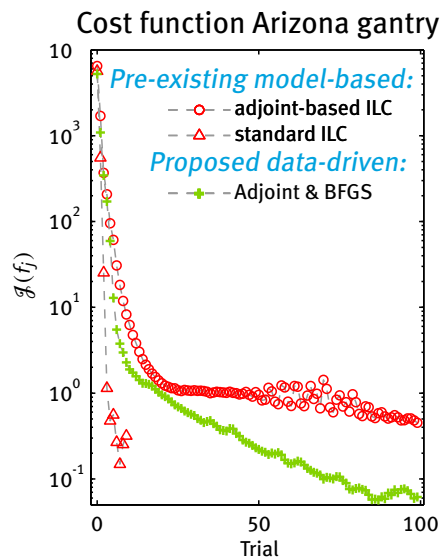
► Gradient-descent ILC algorithm:

$$f_{j+1} = f_j + \alpha J^T e_j$$

► Measure $J^T e_j$ using experiments on J :

$$J^T e_j = \mathcal{T}_{n_i} \left(\sum_{i=1}^{n_i} \sum_{j=1}^{n_o} l_{ij} J l_{ij} \right) \mathcal{T}_{n_o} e_j$$

- Convergence inherently robust for model errors
- Compatible with basis functions



References I

- Bamieh, B. A. & Pearson, J. B. (1992), 'A general framework for linear periodic systems with applications to $\mathcal{H}_\infty$ sampled-data control', *IEEE Transactions on Automatic Control* **37**(4), 418–435.
- Bamieh, B., Pearson, J. B., Francis, B. A. & Tannenbaum, A. (1991), 'A lifting technique for linear periodic systems with applications to sampled-data control', *Systems & Control Letters* **17**(2), 79–88.
- Blanken, L., Boeren, F., Bruijnen, D. & Oomen, T. (2016), Rational feedforward tuning: Approaches, stable inversion, and experimental comparison, in 'Proceedings of the 2016 American Control Conference', Boston, Massachusetts, United States, pp. 2629–2634.
- Blanken, L., de Rozario, R., van Zundert, J., Koekebakker, S., Steinbuch, M. & Oomen, T. (2016), Advanced feedforward and learning control for mechatronic systems, in 'Proceedings of the 3rd DSPE Conference on Precision Mechatronics', Sint-Michielsgestel, The Netherlands, pp. 79–86.
- Boeren, F., Bareja, A., Kok, T. & Oomen, T. (2016), 'Frequency-domain ILC approach for repeating and varying tasks: With application to semiconductor bonding equipment', *IEEE/ASME Transactions on Mechatronics* **21**(6), 2716–2727.
- Boeren, F., Blanken, L., Bruijnen, D. & Oomen, T. (2015), Rational iterative feedforward control: Optimal instrumental variable approach for enhanced performance, in 'Proceedings of the 54th Conference on Decision and Control', Osaka, Japan, pp. 6058–6063.
- Boeren, F., Bruijnen, D., van Dijk, N. & Oomen, T. (2014), 'Joint input shaping and feedforward for point-to-point motion: Automated tuning for an industrial nanopositioning system', *Mechatronics* **24**(6), 572–581.
- Boeren, F., Oomen, T. & Steinbuch, M. (2014), Accuracy aspects in motion feedforward tuning, in 'Proceedings of the 2014 American Control Conference', Portland, Oregon, United States, pp. 2178–2183.
- Boeren, F., Oomen, T. & Steinbuch, M. (2015), 'Iterative motion feedforward tuning: a data-driven approach based on instrumental variable identification', *Control Engineering Practice* **37**, 11–19.
- Bolder, J., Boeren, F. & Oomen, T. (2014), 'Intelligente regeltechniek voor nieuwe generatie mechatronica', *Aandrijftechniek* **10**, 22–24.
- Bolder, J., Kleinendorst, S. & Oomen, T. (2018), 'Data-driven multivariable ILC: Enhanced performance by eliminating L and Q filters', *International Journal of Robust and Nonlinear Control* **28**(12), 3728–3751.
- Bolder, J., Lemmen, B., Koekebakker, S., Oomen, T., Bosgra, O. & Steinbuch, M. (2012), Iterative learning control with basis functions for media positioning in scanning inkjet printers, in 'Proceedings of the 2012 Multi-conference on Systems and Control', Dubrovnik, Croatia, pp. 1255–1260.
- Bolder, J. & Oomen, T. (2015), 'Rational basis functions in iterative learning control - with experimental verification on a motion system', *IEEE Transactions on Control Systems Technology* **23**(2), 722–729.
- Bolder, J. & Oomen, T. (2016), 'Inferential iterative learning control: A 2D-system approach', *Automatica* **71**, 247–253.

- Bolder, J., Oomen, T. & Steinbuch, M. (2014), On inferential iterative learning control: with example on a printing system, in 'Proceedings of the 2014 American Control Conference', Portland, Oregon, United States, pp. 1827–1832.
- Bristow, D. A., Tharayil, M. & Alleyne, A. G. (2006), 'A survey of iterative learning control: A learning-based method for high-performance tracking control', *IEEE Control Systems Magazine* **26**(3), 96–114.
- van der Meulen, S., Tousain, R. L. & Bosgra, O. H. (2008), 'Fixed structure feedforward controller design exploiting iterative trials: Application to a wafer stage and a desktop printer', *Journal of Dynamic Systems, Measurements, and Control* **130**, 051006–1.
- Oomen, T., Grassens, E. & Hendriks, F. (2015), 'Inferential motion control: An identification and robust control framework for unmeasured performance variables', *IEEE Transactions on Control Systems Technology* **23**(4), 1602–1610.
- Oomen, T. & Rojas, C. R. (2017), 'Sparse iterative learning control with application to a wafer stage: Achieving performance, resource efficiency, and task flexibility', *Mechatronics* **47**, 134–137.
- Oomen, T., van de Wijdeven, J. & Bosgra, O. (2009), 'Suppressing intersample behavior in iterative learning control', *Automatica* **45**(4), 981–988.
- Oomen, T., van de Wijdeven, J. & Bosgra, O. (2011), 'System identification and low-order optimal control of intersample behavior in ILC', *IEEE Transactions on Automatic Control* **56**(11), 2734–2739.
- Oomen, T., van der Maas, R., Rojas, C. R. & Hjalmarsson, H. (2014), 'Iterative data-driven $\mathcal{H}_\infty$ norm estimation of multivariable systems with application to robust active vibration isolation', *IEEE Transactions on Control Systems Technology* **22**(6), 2247–2260.
- Rojas, C. R., Oomen, T., Hjalmarsson, H. & Wahlberg, B. (2012), 'Analyzing iterations in identification with application to nonparametric $\mathcal{H}_\infty$ -norm estimation', *Automatica* **48**(11), 2776–2790.
- de Roover, D., Bosgra, O. H. & Steinbuch, M. (2000), 'Internal-model-based design of repetitive and iterative learning controllers for linear multivariable systems', *International Journal of Control* **73**(10), 914–929.
- de Rozario, R. (2015), Identifying position-dependent mechanical systems: A physics-based LPV approach, with application to a wafer stage, Master's thesis, Eindhoven University of Technology, Eindhoven, The Netherlands.
- Smits, P. (2016), Position-dependent feedforward for rational systems: A joint input shaping approach with application to a prototype mechanical system, Master's thesis, Eindhoven University of Technology, Eindhoven, The Netherlands.
- Steinbuch, M., van Berkel, K., Leenknecht, G., Oomen, T. & van de Wijdeven, J. (2009), Reading of cracked optical discs using iterative learning control, in 'Proceedings of the 2009 American Control Conference', St. Louis, Missouri, United States, pp. 258–263.
- Steinbuch, M., van de Wijdeven, J., Oomen, T., van Berkel, K. & Leenknecht, G. (2009), Recovering data from cracked optical discs using Hankel iterative learning control, in P. M. J. Van den Hof, C. Scherer & P. S. C. Heuberger, eds, 'Model-Based Control: Bridging Rigorous Theory and Advanced Technology', Springer, New York, New York, United States, pp. 147–166.
- van de Wijdeven, J. (2008), Iterative Learning Control Design for Uncertain and Time-Windowed Systems, PhD thesis, Eindhoven University of Technology, Eindhoven, The Netherlands.
- van de Wijdeven, J. & Bosgra, O. (2008), 'Residual vibration suppression using Hankel iterative learning control', *International Journal of Robust and Nonlinear Control* **18**(10), 1034–1051.
- van de Wijdeven, J. & Bosgra, O. (2010), 'Using basis functions in iterative learning control: Analysis and design theory', *International Journal of Control* **83**(4), 661–675.
- van Zundert, J., Bolder, J., Koekebakker, S. & Oomen, T. (2016), 'Resource-efficient ILC for LTI/LTV systems through LQ tracking and stable inversion: Enabling large tasks on a position-dependent industrial printer', *Mechatronics* **38**, 76–90.